

Mindscape: Multi-Perspective VR Storytelling in Interactive Psychological Thriller Experiences

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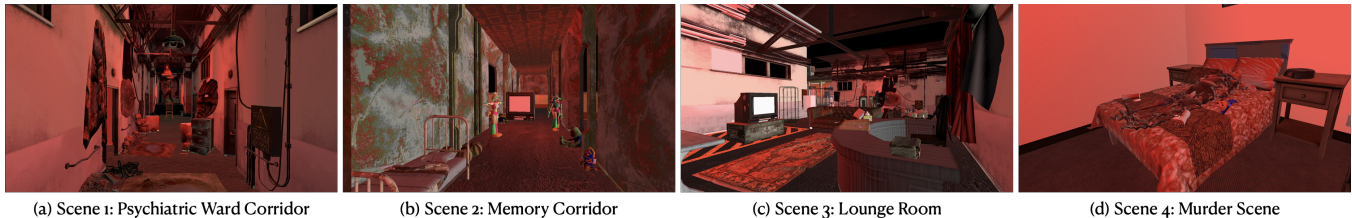


Figure 1: Four main scenes in *Mindscape*.

Abstract

This paper presents *Mindscape*, an immersive virtual reality experience that explores psychological thriller narratives through multiple personality perspectives. Inspired by James Mangold's film "Identity" (2003), the project employs environmental storytelling, perspective-switching mechanics, and symbolic interaction design to create a nonlinear narrative in which players gradually discover their role as both investigator and perpetrator. Through a user study with 10 participants, we identified critical usability challenges, including motion sickness from visual effects, text readability issues, and varying comprehension of psychological metaphors between technical and general audiences. The study contributes insights

for designing accessible VR experiences that handle complex psychological themes while maintaining user comfort and narrative immersion.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *User studies*.

Keywords

Virtual Reality (VR); environmental storytelling; ; interactive narrative; narrative fragmentation

ACM Reference Format:

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1 Introduction

The intersection of psychological thriller narratives and VR offers unique immersive storytelling opportunities that traditional media cannot match [Tomé et al. 2024]. James Mangold's "Identity"

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(2003) [ide 2003] exemplifies sophisticated multi-perspective narrative construction, in which fragmented personalities converge toward a shocking revelation of a unified identity. Its exploration of dissociative identity disorder through spatial metaphors and psychological symbolism provides a foundation for VR adaptation.

Our work, *Mindscape*, translates these cinematic concepts into an interactive experience where players embody a police-appointed psychiatrist investigating serial murders. By switching between personality perspectives, players reconstruct fragmented memories as they gradually discover their complicity. The boundary between reality and psychological projection dissolves as evidence accumulates, culminating in the revelation that the investigator is the perpetrator. Our contributions include: 1) novel interaction paradigms for personality-driven narrative progression, 2) environmental design enhancing psychological immersion in VR, and 3) empirical validation of multi-perspective storytelling effectiveness in virtual environments.

2 Project Overview of *Mindscape*

Our **visual design** constructs spaces that embody fragmented mental states through environmental symbolism, from Identity’s psychological architecture. Psychiatric Ward Corridor (Figure 1a) employs oppressive spatial ratios with decaying white walls and blood-red accents. Industrial lights project unstable illumination, representing mental instability, while embedded facial sculptures and fractured masks symbolize personality fragmentation. Memory Corridor (Figure 1b) features corroded green-gray walls with deep red lighting, symbolizing traumatic suppression through scattered beds, dolls, and distorted figures evoking corrupted childhood memories. Lounge Room (Figure 1c) blends motel and office aesthetics with warm red lighting contrasted by cool localized illumination, creating visual tension that reflects psychological conflicts and serves as a narrative hub for personality transitions. Murder Scene (Figure 1d) depicts a locked-room murder with numbered clues and strategic prop placement, using warm red lighting and high-saturation fabrics to produce visual oppression.

The **interaction design** of *Mindscape* follows the core principle of “narrative-driven interaction [Spaulding and Faste 2013],” in which all trigger events are initiated by the player’s spatial exploration and object engagement, closely linking interactive behavior to the progression of multiple storylines. Beyond standard VR interactions, we implemented advanced narrative mechanics through a multi-layered trigger system in which environmental elements serve as both physical objects and psychological metaphors.

For example, the floating skull in scene 1 levitates when touched, metaphorically representing disordered consciousness in personality dissociation and symbolizing latent connections among multiple identities. Distributed doll trigger (Figure 2a) in the memory corridor (scene 2) activates unique audio-visual responses corresponding to distinct personality traits, while television (Figure 2b) proximity automatically initiates distorted childhood sequences with intensified low-frequency audio, creating psychological pressure. This passive trigger design ensures that players naturally encounter a key narrative node during exploration, deepening their understanding of the origins of personality.



(a) Trigger: Doll (b) Trigger: Television

Figure 2: Interaction design in *Mindscape*.

We also incorporated a document interaction element in the lounge room (scene 3) that uses progressive-revelation mechanics. Approaching objects triggers perspective-specific footage and the emergence of hidden text, including the automatic unfolding of paper that reveals personality profiles. This layered interaction design embeds the “exploration–discovery–information integration” process into the scene, allowing players to actively construct the link between personalities and the storyline. Symbolic evidence collection uses numbered keys and lighting focal shifts to establish identity markers in scene 4, enabling players to construct narrative connections through physical clue-gathering that directly influences subsequent event interpretation, transforming exploration into the primary driving force of narrative reconstruction.

The **technical implementation** of *Mindscape* was developed in Unity 2022.3.17f1c1 using physics systems, animation state machines, and custom shaders for VR interactivity. The floating skull effect uses Unity’s PhysX engine in Custom Force mode with C# scripts adjusting Rigidbody parameters and particle systems, creating light trails. Proximity detection triggers Animation Controller state machines when players enter interaction zones, while HDRP Post-processing creates distorted television effects. Interactive objects use the XR Interaction Toolkit for grab mechanics, and Signed Distance Field Shaders ensure VR text is legible. The system employs Scriptable Object architecture to separate narrative logic from physics interactions, optimizing performance through modular design and synchronized audio-visual event triggering.

3 User Study

We conducted exploratory testing with 10 participants (5 developers, 5 volunteers) using open exploration to observe natural behaviors. Behavioral logs tracked spatial movement, interaction efficiency, engagement patterns, and physiological responses, followed by structured interviews on environmental perception, metaphor interpretation, interaction expectations, discomfort levels, narrative comprehension, and content sensitivity. Key findings: 4 participants experienced dizziness from TV flickering, 5 failed to read 8pt text (requiring $3.2 \times$ longer), and metaphor recognition varied significantly between developers (all 5) and volunteers (only 2 recognized floating skull symbolism). We implemented three optimizations: reduced flickering to 3Hz with audio warnings, added gray luminance to interactive objects, and standardized text to ≥ 14 pt with voice narration.

4 Conclusion and Futurework

This paper demonstrates the potential of VR technology for creating immersive psychological narratives through our escape rooms,

which focus on dissociative identity disorder [Calvert and Hume 2022]. The custom physics-based interactions and multi-layered trigger systems effectively translate complex psychological concepts into engaging mechanics. A user study with 10 participants confirmed strong narrative comprehension and immersion, although improvements in visual cue clarity and content sensitivity options were identified. Our future work will expand narrative complexity, integrate biometric feedback for dynamic environmental adaptation, enhance haptic and audio spatialization, and develop customizable sensitivity settings to support educational and clinical mental health applications.

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